

Library API

REST API Documentation

Version 1.0 • Laravel 11 • Sanctum Authentication
Base URL: http://library-api.test/api
June 2026

Framework	Laravel 11
Auth	Laravel Sanctum — Bearer Token
Format	JSON — Content-Type: application/json
Database	MySQL
Tests	43 tests, 119 assertions — all passing
User Types	Staff (admin / librarian) and Member
Scheduler	app:expire-reservations runs hourly — auto-expires unclaimed reservations

Authentication

All protected endpoints require:
Authorization: Bearer YOUR_TOKEN

Staff token: POST /api/staff/login
Member token: POST /api/member/login

Auth-Staff

POST

StaffLogin

Public

<http://library-api.test/api/staff/login>

No authentication required — public endpoint

REQUEST BODY

Staff credentials — email and password

```
{
  "email": "admin@library.com",
  "password": "password"
}
```

EXAMPLE RESPONSE

```
{
  "message": "Login successful.",
  "token": "2|abc123xyz...",
  "data": {
    "id": 1,
    "name": "Admin User",
    "email": "admin@library.com",
    "role": "admin"
  }
}
```

GET

Get Staff Profile

Staff Token

<http://library-api.test/api/staff/me>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "data": {
    "id": 1,
    "name": "Admin User",
    "email": "admin@library.com",
    "role": "admin"
  }
}
```

POST

Staff Logout

Staff Token

<http://library-api.test/api/staff/logout>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "message": "Logged out successfully."
}
```

Auth-Member

POST

MemberLogin

Public

http://library-api.test/api/member/login

No authentication required — public endpoint

REQUEST BODY

Member credentials — email and password

```
{
  "email": "ade@member.com",
  "password": "password"
}
```

EXAMPLE RESPONSE

```
{
  "message": "Login successful.",
  "token": "5|xyz789abc...",
  "data": {
    "id": 1,
    "first_name": "Ade",
    "last_name": "Regular",
    "email": "ade@member.com"
  }
}
```

GET

Get Member Profile

Member Token

http://library-api.test/api/member/me

Member token required

EXAMPLE RESPONSE

```
{
  "data": {
    "id": 1,
    "first_name": "Ade",
    "last_name": "Regular",
    "email": "ade@member.com",
    "status": "active",
    "member_tier": "Regular"
  }
}
```

POST

Member Logout

Member Token

http://library-api.test/api/member/logout

Member token required

EXAMPLE RESPONSE

```
{
  "message": "Logged out successfully."
}
```

Catalog - Categories

GET

List Categories

Public

<http://library-api.test/api/categories>

No authentication required — public endpoint

EXAMPLE RESPONSE

```
{
  "data": [
    {"id": 1, "name": "Fiction", "description": "Fictional literature"},
    {"id": 2, "name": "Non-Fiction", "description": "Non-fictional works"}
  ]
}
```

POST

Create Categories

Staff Token

<http://library-api.test/api/categories>

Staff token required — librarian or admin

REQUEST BODY

```
{
  "name": "Biography",
  "description": "Biographical works about real people"
}
```

EXAMPLE RESPONSE

```
{
  "message": "Category created successfully.",
  "data": {"id": 3, "name": "Biography", "description": "Biographical works about real people"}
}
```

Catalog - Authors

GET

List Authors

Public

<http://library-api.test/api/authors>

No authentication required — public endpoint

EXAMPLE RESPONSE

```
{
  "data": [
    {"id": 1, "name": "Chinua Achebe", "bio": "Nigerian novelist and poet."},
    {"id": 2, "name": "Wole Soyinka", "bio": "Nigerian playwright and poet."}
  ]
}
```

POST

Create Authors

Staff Token

http://library-api.test/api/authors

Staff token required — librarian or admin

REQUEST BODY

```
{
  "name": "Ben Okri",
  "bio": "Nigerian poet and novelist, winner of the Booker Prize."
}
```

EXAMPLE RESPONSE

```
{
  "message": "Author created successfully.",
  "data": {"id": 4, "name": "Ben Okri", "bio": "Nigerian poet and novelist."}
}
```

PATCH

Update Authors

Staff Token

http://library-api.test/api/authors/1

Staff token required — librarian or admin

REQUEST BODY

All fields optional — send only what you want to update

```
{
  "bio": "Nigerian novelist, poet, and professor. Author of Things Fall Apart."
}
```

EXAMPLE RESPONSE

```
{
  "message": "Author updated successfully.",
  "data": {"id": 1, "name": "Chinua Achebe", "bio": "Nigerian novelist, poet, and professor."}
}
```

Catalog - Books

GET

List Books

Public

<http://library-api.test/api/books>

No authentication required — public endpoint

EXAMPLE RESPONSE

```
{
  "data": [
    {
      "id": 1,
      "title": "Things Fall Apart",
      "isbn": "978-0-385-47454-2",
      "published_year": 1958,
      "is_retired": false,
      "authors": [{"name": "Chinua Achebe"}],
      "categories": [{"name": "Fiction"}],
      "available_copies": 2
    }
  ]
}
```

GET

Filter Books by Title

Public

<http://library-api.test/api/books?title=Things>

No authentication required — public endpoint

EXAMPLE RESPONSE

```
{
  "data": [
    {"id": 1, "title": "Things Fall Apart", "isbn": "978-0-385-47454-2"}
  ]
}
```

GET

Get Single Book

Public

<http://library-api.test/api/books/1>

No authentication required — public endpoint

EXAMPLE RESPONSE

```
{
  "data": {
    "id": 1,
    "title": "Things Fall Apart",
    "isbn": "978-0-385-47454-2",
    "description": "A classic Nigerian novel.",
    "published_year": 1958,
    "authors": [{"name": "Chinua Achebe"}],
    "categories": [{"name": "Fiction"}],
    "copies": [
      {"copy_number": 1, "condition": "good", "status": "available"},
      {"copy_number": 2, "condition": "good", "status": "available"}
    ]
  }
}
```

POST

Create Book

Staff Token

<http://library-api.test/api/books>

Staff token required — librarian or admin

REQUEST BODY

copies is an integer — number of copies to create

```
{
  "title": "Arrow of God",
  "isbn": "978-0-385-01480-0",
  "description": "Third novel in Chinua Achebe's African Trilogy.",
  "published_year": 1964,
  "author_ids": [1],
  "category_ids": [1],
  "copies": 2
}
```

EXAMPLE RESPONSE

```
{
  "message": "Book created successfully.",
  "data": {
    "id": 4,
    "title": "Arrow of God",
    "isbn": "978-0-385-01480-0",
    "published_year": 1964,
    "authors": [{ "name": "Chinua Achebe" }],
    "categories": [{ "name": "Fiction" }]
  }
}
```

PATCH

Update Book

Staff Token

<http://library-api.test/api/books/1>

Staff token required — librarian or admin

REQUEST BODY

All fields optional — send only what you want to update

```
{
  "description": "Chinua Achebe's debut novel, widely regarded as the finest African novel ever written."
}
```

EXAMPLE RESPONSE

```
{
  "message": "Book updated successfully.",
  "data": { "id": 1, "title": "Things Fall Apart", "description": "Chinua Achebe's debut novel..." }
}
```

DELETE

Retire Book

Staff Token

<http://library-api.test/api/books/3>
Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "message": "Book retired successfully."
}
```

BUSINESS RULES

- Cannot retire a book with active loans — returns 409 Conflict
- Retired books are hidden from the public catalog
- Staff with is_retired filter can still view retired books

ERROR RESPONSES

Status	Message
409	Book has active loans and cannot be retired.

GET

Filter Book by Author

Public

<http://library-api.test/api/books?author=chinua>
No authentication required — public endpoint

EXAMPLE RESPONSE

```
{
  "data": [
    {
      "id": 1,
      "title": "Things Fall Apart",
      "authors": [{"name": "Chinua Achebe"}]
    }
  ]
}
```

GET

Filter Book By Category

Public

<http://library-api.test/api/books?category=Fiction>
No authentication required — public endpoint

EXAMPLE RESPONSE

```
{
  "data": [
    {
      "id": 1,
      "title": "Things Fall Apart",
      "categories": [{"name": "Fiction"}]
    }
  ]
}
```

Loans

POST

Borrow Book

Member Token

http://library-api.test/api/loans

Member token required

REQUEST BODY

Member must be active with no unpaid fines and within tier book limit

```
{
  "book_id": 1
}
```

EXAMPLE RESPONSE

```
{
  "message": "Book borrowed successfully.",
  "data": {
    "id": 1,
    "status": "active",
    "borrowed_at": "2026-06-22 14:50:54",
    "due_date": "2026-07-06 14:50:54",
    "returned_at": null,
    "renewals_count": 0,
    "fines_accrued": "0.00",
    "is_overdue": false,
    "book": {"title": "Things Fall Apart", "isbn": "978-0-385-47454-2"},
    "member": {"first_name": "Ade", "last_name": "Regular", "email": "ade@member.com"}
  }
}
```

BUSINESS RULES

- Member must be active — suspended members receive 403
- Member must have zero unpaid fines — any unpaid fine blocks borrowing
- Member cannot exceed their tier max_books limit (Regular: 3, Premium: 6)
- Member cannot borrow the same book twice simultaneously
- Only non-damaged, available copies are assigned
- If a notified reservation with an active claim window exists, borrowing is blocked

ERROR RESPONSES

Status	Message
403	Member account is suspended.
403	Member has unpaid fines.
422	Member has reached their borrowing limit.
422	Member already has an active loan for this book.
422	No available copy found for this book.

GET

List Loans

Member Token

http://library-api.test/api/loans/1

Member token required

EXAMPLE RESPONSE

```
{
  "data": [
    {
      "id": 1,
      "status": "active",
      "borrowed_at": "2026-06-22 14:50:54",
      "due_date": "2026-07-06 14:50:54",
      "is_overdue": false,
      "book": {"title": "Things Fall Apart"}
    }
  ],
  "meta": {"current_page": 1, "last_page": 1, "per_page": 15, "total": 1}
}
```

PATCH

Renew Loan

Member Token

http://library-api.test/api/loans/1/renew

Member token required

EXAMPLE RESPONSE

```
{
  "message": "Loan renewed successfully.",
  "data": {
    "id": 1,
    "status": "active",
    "due_date": "2026-07-20 14:50:54",
    "renewals_count": 1
  }
}
```

BUSINESS RULES

- Maximum 2 renewals per loan — enforced by MAX_RENEWALS constant
- Extends due_date by the member tier loan_period_days from current due date
- Loan must have status: active to be renewed

ERROR RESPONSES

Status	Message
--------	---------

422	Maximum renewals reached.
-----	---------------------------

403	Only active loans can be renewed.
-----	-----------------------------------

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Return Loan

Staff Token

http://library-api.test/api/loans/1/return

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "message": "Book returned successfully.",
  "data": {
    "id": 1,
    "status": "returned",
    "borrowed_at": "2026-06-22 14:50:54",
    "due_date": "2026-07-06 14:50:54",
    "returned_at": "2026-06-22 15:30:00",
    "renewals_count": 0,
    "fines_accrued": "0.00",
    "is_overdue": false
  }
}
```

BUSINESS RULES

- Staff only — librarian or admin
- If returned after due_date, a fine is automatically created
- Fine amount = days overdue x member tier fine_rate (default N50/day)
- If total unpaid fines exceed N5,000 the member is automatically suspended
- On return, the system checks for pending reservations for this book
- If a pending reservation exists, the first in queue is notified with a 48-hour claim window

ERROR RESPONSES

Status	Message
404	Loan not found.

Reservations

POST

Create Reservation

Member Token

http://library-api.test/api/reservations

Member token required

REQUEST BODY

Joins FIFO queue — queue_position assigned automatically

```
{
  "book_id": 2
}
```

EXAMPLE RESPONSE

```
{
  "message": "Reservation created successfully.",
  "data": {
    "id": 1,
    "status": "pending",
    "queue_position": 1,
    "reserved_at": "2026-06-22 15:03:10",
    "notified_at": null,
    "claim_expires_at": null,
    "is_claim_expired": false
  }
}
```

BUSINESS RULES

- FIFO queue — queue_position assigned automatically based on existing reservations
- When notified, member has 48 hours to borrow the book (claim_expires_at)
- If claim expires unclaimed, the next member in queue is notified automatically
- Expired reservations are processed hourly by the scheduler

ERROR RESPONSES

Status	Message
--------	---------

422	Member already has a reservation for this book.
-----	---

GET

List Reservation

Member Token

http://library-api.test/api/reservations

Member token required

EXAMPLE RESPONSE

```
{
  "data": [
    {
      "id": 1,
      "status": "pending",
      "queue_position": 1,
      "reserved_at": "2026-06-22 15:03:10",
      "notified_at": null,
      "claim_expires_at": null
    }
  ]
}
```

DELETE

Cancel Reservation

Member Token

<http://library-api.test/api/reservations/1>

Member token required

EXAMPLE RESPONSE

```
{
  "message": "Reservation cancelled successfully."
}
```

BUSINESS RULES

- Member can only cancel their own reservation
- Cancelling does not automatically notify the next member in queue

Fines

GET

List Fines

Member Token

<http://library-api.test/api/fines>

Member token required

EXAMPLE RESPONSE

```
{
  "data": [
    {
      "id": 1,
      "amount": "150.00",
      "type": "overdue",
      "is_paid": false,
      "paid_at": null,
      "loan_id": 1
    }
  ]
}
```

PATCH

Pay Fine

Staff Token

<http://library-api.test/api/fines/1/pay>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "message": "Fine marked as paid.",
  "data": {
    "id": 1,
    "amount": "150.00",
    "is_paid": true,
    "paid_at": "2026-06-22 16:00:00"
  }
}
```

BUSINESS RULES

- Staff only — marks the fine as paid and records paid_at timestamp

Members

POST

Register Member

Public

<http://library-api.test/api/members>

No authentication required — public endpoint

REQUEST BODY

member_tier_id: 1 = Regular, 2 = Premium

```
{
  "first_name": "Chidi",
  "last_name": "Okeke",
  "email": "chidi@member.com",
  "password": "password",
  "password_confirmation": "password",
  "phone_number": "08011112222",
  "member_tier_id": 1
}
```

EXAMPLE RESPONSE

```
{
  "message": "Member registered successfully.",
  "data": {
    "id": 3,
    "first_name": "Chidi",
    "last_name": "Okeke",
    "email": "chidi@member.com",
    "status": "active",
    "member_tier": "Regular"
  }
}
```

GET

List Members

Staff Token

<http://library-api.test/api/members>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "data": [
    {"id": 1, "first_name": "Ade", "last_name": "Regular", "email": "ade@member.com", "status": "active"},
    {"id": 2, "first_name": "Bola", "last_name": "Premium", "email": "bola@member.com", "status": "active"}
  ],
  "meta": {"current_page": 1, "last_page": 1, "per_page": 15, "total": 2}
}
```

GET

Get Single Member

Staff Token

<http://library-api.test/api/members/1>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "data": {
    "id": 1,
    "first_name": "Ade",
    "last_name": "Regular",
    "email": "ade@member.com",
    "phone_number": "08012345678",
    "status": "active",
    "member_tier": "Regular",
    "joined_at": "2026-06-22"
  }
}
```

PATCH

Update Member

Member Token

<http://library-api.test/api/members/1>

Member token required

REQUEST BODY

Member can only update their own profile — staff cannot use this endpoint

```
{
  "phone_number": "08099999999"
}
```

EXAMPLE RESPONSE

```
{
  "message": "Profile updated successfully.",
  "data": {"id": 1, "first_name": "Ade", "phone_number": "08099999999"}
}
```

PATCH

Suspend Member

Staff Token

<http://library-api.test/api/members/2/suspend>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "message": "Member suspended successfully.",
  "data": {"id": 2, "first_name": "Bola", "status": "suspended"}
}
```

BUSINESS RULES

- Manual suspension by staff — independent of fine threshold

PATCH

Reinstate Member

Staff Token

<http://library-api.test/api/members/2/reinstate>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "message": "Member reinstated successfully.",
  "data": {"id": 2, "first_name": "Bola", "status": "active"}
}
```

BUSINESS RULES

- Restores member status to active — clears suspension regardless of fine balance

PATCH

Upgrade Member Tier

Staff Token

<http://library-api.test/api/members/1/upgrade-tier>

Staff token required — librarian or admin

REQUEST BODY

Staff only — 1 = Regular, 2 = Premium

```
{
  "member_tier_id": 2
}
```

EXAMPLE RESPONSE

```
{
  "message": "Member tier upgraded successfully.",
  "data": {"id": 1, "first_name": "Ade", "member_tier": "Premium"}
}
```

BUSINESS RULES

- Regular: 3 books max, 14-day loan | Premium: 6 books max, 21-day loan

Member Tiers

GET

List Member Tiers

Public

<http://library-api.test/api/member-tiers>

No authentication required — public endpoint

EXAMPLE RESPONSE

```
{
  "data": [
    {"id": 1, "name": "Regular", "max_books": 3, "loan_period_days": 14, "fine_rate": "50.00"},
    {"id": 2, "name": "Premium", "max_books": 6, "loan_period_days": 21, "fine_rate": "50.00"}
  ]
}
```


POST

Create Tier

Staff Token

<http://library-api.test/api/member-tiers>

Staff token required — librarian or admin

REQUEST BODY

Admin only — *fine_rate* is in Naira per day

```
{
  "name": "Student",
  "max_books": 2,
  "loan_period_days": 7,
  "fine_rate": 50
}
```

EXAMPLE RESPONSE

```
{
  "message": "Member tier created successfully.",
  "data": {"id": 3, "name": "Student", "max_books": 2, "loan_period_days": 7, "fine_rate": "50.00"}
}
```

PATCH

Update Tier

Staff Token

<http://library-api.test/api/member-tiers/1>

Staff token required — librarian or admin

REQUEST BODY

Admin only — *all fields optional*

```
{
  "max_books": 4
}
```

EXAMPLE RESPONSE

```
{
  "message": "Member tier updated successfully.",
  "data": {"id": 1, "name": "Regular", "max_books": 4}
}
```

Staff Management

GET

List Staff

Staff Token

<http://library-api.test/api/staff>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "data": [
    {"id": 1, "name": "Admin User", "email": "admin@library.com", "role": "admin"},
    {"id": 2, "name": "Librarian User", "email": "librarian@library.com", "role": "librarian"}
  ]
}
```

POST

Create Staff

Staff Token

http://library-api.test/api/staff

Staff token required — librarian or admin

REQUEST BODY

Admin only — role must be: librarian or admin

```
{
  "name": "New Librarian",
  "email": "newlibrarian@library.com",
  "password": "password",
  "password_confirmation": "password",
  "role": "librarian"
}
```

EXAMPLE RESPONSE

```
{
  "message": "Staff created successfully.",
  "data": {"id": 3, "name": "New Librarian", "email": "newlibrarian@library.com", "role": "librarian"}
}
```

PATCH

Update Staff

Staff Token

http://library-api.test/api/staff/3

Staff token required — librarian or admin

REQUEST BODY

Admin only — all fields optional

```
{
  "name": "Updated Librarian"
}
```

EXAMPLE RESPONSE

```
{
  "message": "Staff updated successfully.",
  "data": {"id": 3, "name": "Updated Librarian", "role": "librarian"}
}
```

DELETE

Delete Staff

Staff Token

http://library-api.test/api/staff/3

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "message": "Staff deleted successfully."
}
```

Reports

GET

Dashboard

Staff Token

<http://library-api.test/api/reports/dashboard>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "data": {
    "total_books": 4,
    "total_members": 3,
    "active_loans": 1,
    "overdue_loans": 0,
    "total_fines_unpaid": "0.00"
  }
}
```

GET

Overdue Loans

Staff Token

<http://library-api.test/api/reports/overdue>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "data": [
    {
      "id": 2,
      "member": {"first_name": "Ade", "email": "ade@member.com"},
      "book": {"title": "Things Fall Apart"},
      "due_date": "2026-06-01",
      "days_overdue": 21
    }
  ]
}
```

GET

Low Stock

Staff Token

<http://library-api.test/api/reports/low-stock>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "data": [
    {
      "id": 1,
      "title": "Things Fall Apart",
      "total_copies": 2,
      "available_copies": 0
    }
  ]
}
```

GET

Most Borrowed

Staff Token

<http://library-api.test/api/reports/most-borrowed>

Staff token required — librarian or admin

EXAMPLE RESPONSE

```
{
  "data": [
    {"id": 1, "title": "Things Fall Apart", "borrow_count": 5},
    {"id": 2, "title": "Purple Hibiscus", "borrow_count": 3}
  ]
}
```

Business Rules Summary

Member Tiers

- Regular: max 3 books, 14-day loan period, N50/day fine rate
- Premium: max 6 books, 21-day loan period, N50/day fine rate

Fines

- Fine = days overdue x tier fine_rate
- Any unpaid fine blocks new loans
- Total unpaid fines exceeding N5,000 auto-suspends the member

Loans

- Maximum 2 renewals per loan
- A member cannot borrow the same book twice simultaneously
- Only non-damaged, available copies are assigned

Reservations

- FIFO queue — queue_position assigned in order of creation
- On book return, the first pending reservation is notified automatically
- Notified member has 48 hours to borrow (claim_expires_at)
- If unclaimed, reservation is marked expired and next in queue is notified
- Expiry check runs every hour via the Laravel scheduler

Books

- Books with active loans cannot be deleted — returns 409
- Deleted books are marked retired and hidden from public catalog

Suspension

- Manual suspension: staff can suspend any member at any time
- Auto-suspension: triggers when unpaid fines exceed N5,000
- Reinstatement is always manual — staff must call the reinstate endpoint